Uneven Access to Neurodiversity Services Across Chicago

Early developmental services are critical for neurodivergent children, yet access may depend on neighborhood socioeconomic conditions. This analysis examines whether service availability is distributed equitably across Chicago's diverse communities and identifies gaps that could guide resource allocation.

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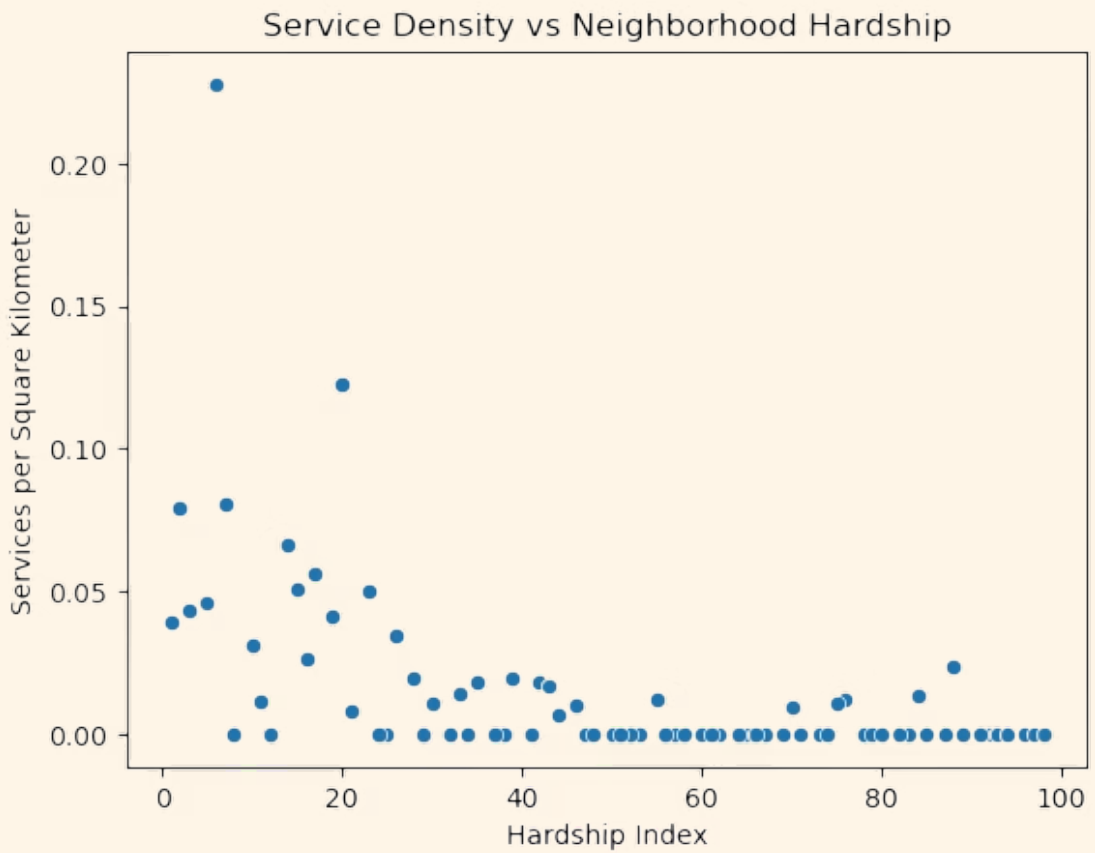
Service Availability Varies Dramatically by Neighborhood Hardship



High Concentration

Low-hardship neighborhoods have the highest service density, creating service-rich zones

Extreme Scarcity

High-hardship neighborhoods have extremely limited service availability, with large measured gaps (often >10×) in ND-proxy service density

Clear Pattern

Service density is associated with neighborhood socioeconomic conditions

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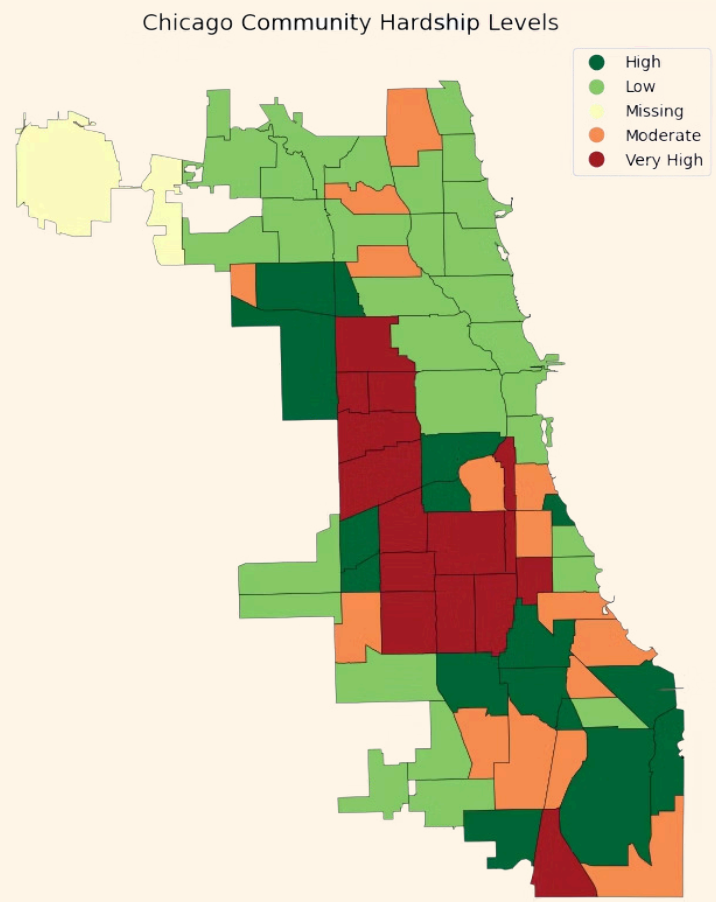
Service Deserts Concentrated in Underinvested Neighborhoods



desert_zero	hardship_index	services_per_sq_km_broad	per_capita_income_
0	31.375	0.038436	35199.062500
1	62.400	0.000000	18710.977778

Average Hardship Score

Service desert areas face significantly higher socioeconomic hardship compared to non-desert areas (31)

Median Income

Desert areas have dramatically lower median household income than non-desert areas ($35K)

Services per km²

Service access is nearly nonexistent in desert areas, with density of approximately zero services per square kilometer

Geography

Desert areas disproportionately located to the west and south side of Chicago.

*Desert definitions: Desert Zero = 45/77, Desert Rank25 = 19/77 (modeled target)

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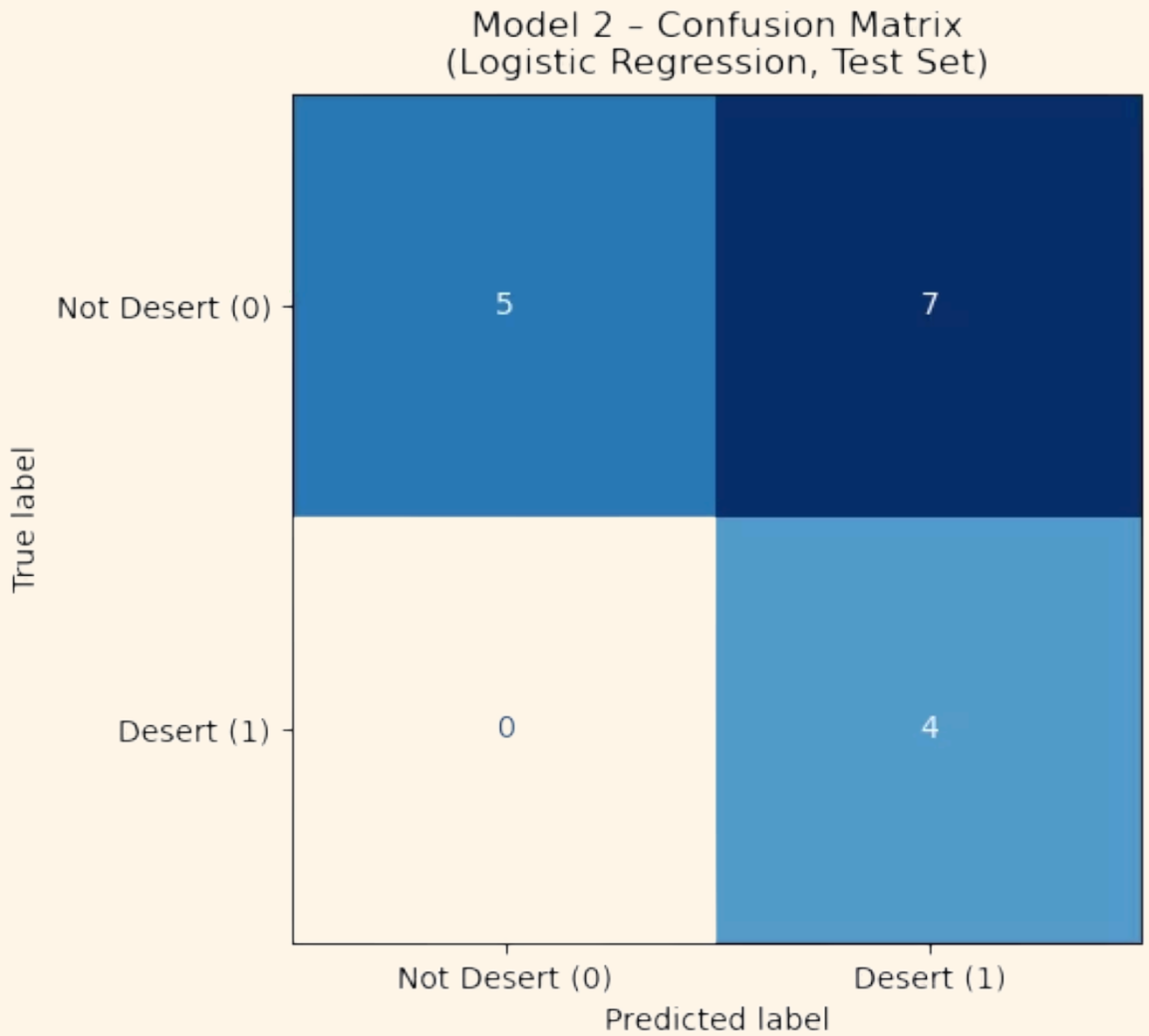
Modeling Confirms Socioeconomic Barriers



Linear Regression Results

Socioeconomic variables explain (**R^2 = 0.227**) about **22.7% of variation** in service density across neighborhoods.

Logistic Regression Results

Practically, the model catches all of the highest-need desert neighborhoods in our test data, but it also flags some extra neighborhoods, so it's best used as an early-warning screen for follow-up rather than a final decision tool.

- Accuracy **0.562**, Precision **0.364**, Recall **1.000**, F1 **0.533**
- Interpretation: **high recall, lower precision**

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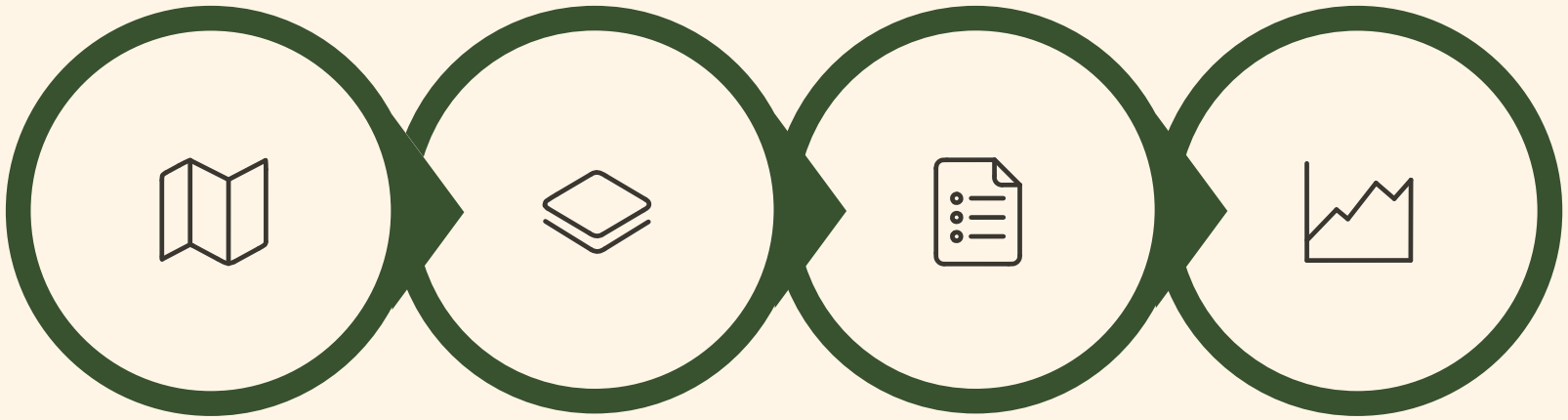
Mapping Services to Neighborhoods



01

OpenStreetMap Services

Geocoded locations of neurodiversity-related services across Chicago (OSM is a proxy*)

02

Spatial Join

Services mapped to Chicago community areas using geographic coordinates

03

Neighborhood Counts

Service counts aggregated by community area boundaries

04

Normalization

Services per square kilometer calculated for cross-neighborhood comparison

*Because OSM is crowd sourced, service listings and categories may not always be complete, and coverage can vary across neighborhoods

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Targeted Policy Action Needed

Identify Gaps

High-hardship neighborhoods show very low measured service density in our proxy dataset

Map Deserts

Service desert identification guides strategic planning and highlights a short list of areas for follow-up validation

Allocate Resources

Targeted funding can address geographic disparities, OSM tagging gaps are themselves actionable through community mapping

This descriptive analysis highlights neighborhoods where service access warrants immediate investigation.

Deliverable: Reproducible pipeline (CSV + SQLite) that can be refreshed as data improves

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